

Multiple Choice (10 marks)

1. Statement 1 | If aH is an element of a factor group, then $|aH|$ divides $|a|$. Statement 2 | If H and K are subgroups of G then HK is a subgroup of G .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. The set of all real numbers under the usual multiplication operation is not a group since
 - (a) multiplication is not a binary operation
 - (b) multiplication is not associative
 - (c) identity element does not exist
 - (d) zero has no inverse
3. Statement 1 | If H is a subgroup of G and a belongs to G then $|aH| = |Ha|$. Statement 2 | If H is a subgroup of G and a and b belong to G , then aH and Hb are identical or disjoint.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
5. Statement 1 | If R is a ring and $f(x)$ and $g(x)$ are in $R[x]$, then $\deg(f(x)+g(x)) = \max(\deg f(x), \deg g(x))$. Statement 2 | If F is a field then every ideal of $F[x]$ is principal.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The characteristic of the ring $2\mathbb{Z}$ is 12, suggesting it has a finite order with respect to addition.
7. True or False: A complete graph with 10 vertices has 45 edges, as each vertex is connected to every other vertex.
8. True or False: A group of order $2p$, where p is an odd prime, is always cyclic and cannot be isomorphic to D_p .
9. True or False: The polynomial $x^2 + 1$ has a zero in the finite field $\mathbb{Z}_2$ at the element 1.
10. True or False: The symmetric group S_3 is a cyclic group, meaning it can be generated by a single element.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the function $f(x,y) = x^2 - 2xy + y^3$ and discuss the significance of its relative extrema, particularly focusing on the line $x = y$. What implications does this have for the function's behavior?
12. Analyze the polynomial function $f(x) = 2x^5 + 8x - 7$ and determine the number of real roots it possesses. Discuss your reasoning and the methods used in your analysis.

End of Paper